1.3 Fractions

Fractions are used to represent parts of a whole. For example, a beam divided into four equal pieces can be represented by

$$\frac{1}{4}$$

Fractions arise frequently in engineering, science and everyday life when a quantity is divided into equal parts.

Definition

A fraction has two parts:

$$\frac{a}{b}$$

where:

- a is the numerator.
- b is the denominator.

The denominator tells us how many equal parts make a whole. The numerator tells us how many of those parts are being considered.

1.3.1 Fractions as Division

A fraction can also be viewed as a division.

$$\frac{a}{b} = a \div b$$

For example,

$$\frac{3}{4} = 3 \div 4 = 0.75$$

Understanding this idea will be important later when studying ratios, gradients and engineering calculations.

1.3.2 Proper and Improper Fractions

Definition

A proper fraction has a numerator smaller than its denominator.

Examples:

$$\frac{1}{2}, \quad \frac{3}{4}, \quad \frac{5}{8}$$

An improper fraction has a numerator greater than or equal to its denominator.
Examples:

$$\frac{5}{4}, \quad \frac{7}{3}, \quad \frac{12}{12}$$

Worked Example

Which of the following fractions are improper?

$$\frac{3}{5}, \quad \frac{7}{4}, \quad \frac{5}{5}, \quad \frac{2}{7}$$

The improper fractions are and

1.3.3 Equivalent Fractions

Definition

Equivalent fractions have the same value.

Multiplying or dividing both the numerator and denominator by the same non-zero number produces an equivalent fraction.

Worked Example

$$\frac{1}{2} =$$

All of these fractions represent the same quantity.

A useful way to think about equivalent fractions is in terms of measurement. Suppose a metal bar has length

$$\frac{1}{2}\ \text{m}$$

If the metre is divided into 2 equal parts, the bar occupies 1 of those parts.

Now divide the same metre into 4 equal parts. The bar occupies 2 of those parts, giving

$$\frac{2}{4}\ \text{m}$$

The physical length of the bar has not changed. Only the size of the units used to describe the length has changed. Similarly,

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$$

Each fraction describes exactly the same quantity.

Equivalent fractions are important because they allow fractions to be expressed using the same denominator. This is similar to converting measurements into the same units before comparing them.

1.3.4 Simplifying Fractions

A fraction is in its simplest form when the numerator and denominator have no common factor other than 1.

Worked Example

Simplify

$$\frac{12}{18}$$

Divide the numerator and denominator by 6 (the highest common factor):

$$\frac{12}{18} =$$

1.3.5 Finding a Common Denominator

When adding or subtracting fractions, the denominators must be the same.

The lowest common denominator (LCD) is the smallest common multiple of the denominators.

Worked Example

Find the LCD of $\frac{1}{3}$ and $\frac{1}{4}$.

Multiples of 3:

$$3, 6, 9, 12, 15, \ldots$$

Multiples of 4:

$$4, 8, 12, 16, \ldots$$

The lowest common denominator is

1.3.6 Adding and Subtracting Fractions

Fractions may be added/subtracted if every denominator is the same

Worked Example

$$\frac{4}{7} + \frac{2}{7} =$$

Worked Example

$$\frac{1}{3} - \frac{2}{3} + \frac{5}{3} =$$

If the denominators are **not** the same, then we must first make them so, by scaling both numerator and denominator of one or more of the fractions by an appropriate amount.

Worked Example

$$\frac{3}{5} - \frac{1}{2} =$$

Worked Example

$$\frac{2}{5}+\frac{7}{15}=$$

1.3.7 Multiplying Fractions

Definition

To multiply fractions:

1. Multiply the numerators.
2. Multiply the denominators.
3. Simplify if possible.

$$\frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}$$

Worked Example

$$\frac{4}{9}\times\frac{3}{5}=$$

1.3.8 Cancelling Before Multiplying

Large calculations can often be simplified before carrying out the multiplication.

Worked Example

Evaluate

$$\frac{3}{4} \times \frac{8}{9}$$

Cancel common factors:

$$\frac{3}{4} \times \frac{8}{9} =$$

Therefore

1.3.9 Dividing Fractions

Division can be interpreted as asking:

"How many of the second quantity fit into the first quantity?"

This interpretation is often useful when working with fractions.

Worked Example

A metal strip has length

$$\frac{3}{4}\ \text{m}$$

It is to be cut into pieces, each of length

$$\frac{1}{4}\ \text{m}$$

How many pieces can be produced?

This calculation is

$$\frac{3}{4} \div \frac{1}{4}$$

The question is simply:

How many quarter-metre lengths fit into three-quarters of a metre?

Since

$$\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

there are three quarter-metre lengths. Therefore,

$$\frac{3}{4} \div \frac{1}{4} = 3$$

This example shows that division of fractions is really a counting process. We are counting how many copies of one fraction fit into another.

For simple fractions this can often be done by inspection. However, for more complicated fractions it is much quicker to use the reciprocal method.

Definition

The reciprocal of a fraction is obtained by exchanging the numerator and denominator. For example,

$$\frac{3}{5} \longrightarrow \frac{5}{3}$$

A useful rule for dividing fractions is:

Definition

$$\frac{a}{b} \div \frac{c}{d} \quad \text{or} \quad \frac{\frac{a}{b}}{\frac{c}{d}} \quad = \quad \frac{a}{b} \times \frac{d}{c} \quad = \quad \frac{ad}{bc}$$

Keep the first fraction, change division to multiplication, and flip the second fraction.

Worked Example

$$\frac{2}{3} \div \frac{5}{7} =$$

1.3.10 Fractions and BIDMAS

Worked Example

$$\frac{1}{2} + \frac{3}{4} \times 2$$

Perform multiplication first:

Then add:

1.3.11 Common Mistakes

- Adding denominators when adding fractions.
- Forgetting to find a common denominator.
- Multiplying the numerator but not the denominator.
- Forgetting to flip the second fraction when dividing.
- Failing to simplify the answer.
- Forgetting that a fraction represents division.

1.3.12 Exercises

Exercise

1. Write an equivalent fraction to $\frac{2}{3}$.
2. Simplify $\frac{15}{25}$.
3. Identify whether $\frac{9}{5}$ is proper or improper.
4. Find the LCD of 3 and 4.
5. $\frac{3}{7} + \frac{2}{7}$
6. $\frac{5}{8} - \frac{1}{8}$
7. $\frac{1}{2} + \frac{1}{4}$
8. $\frac{3}{4} - \frac{1}{2}$
9. $\frac{2}{5} \times \frac{3}{7}$
10. $\frac{4}{9} \times \frac{3}{8}$
11. $\frac{2}{3} \div \frac{4}{5}$
12. $\frac{5}{6} \div \frac{1}{2}$
13. Simplify $\frac{24}{36}$.
14. Convert $\frac{5}{4}$ to a decimal.
15. Convert $\frac{7}{2}$ to a decimal.
16. $\frac{2}{3} + \frac{1}{6}$
17. $\frac{7}{10} - \frac{1}{5}$
18. $\frac{3}{4} \times \frac{2}{3}$
19. $\frac{4}{5} \div \frac{2}{3}$
20. $\frac{5}{4} + \frac{3}{4}$
21. $\frac{9}{4} - \frac{1}{2}$
22. $\frac{1}{2} + \frac{3}{4} \times 2$
23. $\frac{3}{5} \div \frac{9}{10}$
24. $\frac{1}{2} + \frac{3}{4} - \frac{1}{6}$
25. $\left(\frac{3}{4} + \frac{2}{3}\right) \div \frac{5}{6}$

Answers to Exercise 1.3.12

1. $\frac{4}{6}$ or equivalent
2. $\frac{3}{5}$
3. Improper
4. 12
5. $\frac{5}{7}$
6. $\frac{1}{2}$
7. $\frac{3}{4}$
8. $\frac{1}{4}$
9. $\frac{6}{35}$
10. $\frac{1}{6}$
11. $\frac{5}{6}$
12. $\frac{5}{3}$
13. $\frac{2}{3}$
14. 1.25
15. 3.5
16. $\frac{5}{6}$
17. $\frac{1}{2}$
18. $\frac{1}{2}$
19. $\frac{6}{5}$
20. 2
21. $\frac{7}{4}$
22. 2
23. $\frac{2}{3}$
24. $\frac{13}{12}$
25. $\frac{17}{10}$